

Geometric spring-force correction (meeting note)

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1 What the feedback means (one sentence)

A physical spring force should act along the *current line connecting its endpoints* (the deformed chord), but our previous write-up/code used the *relative displacement* vector to set the force direction, which can point in a different direction under shear-like motion.

2 The key distinction

Let a bond connect nodes A and B .

Reference (undeformed) geometry:

$$\mathbf{X}_A^{eq}, \mathbf{X}_B^{eq}, \quad \mathbf{R}_0 = \mathbf{X}_B^{eq} - \mathbf{X}_A^{eq}, \quad L_0 = \|\mathbf{R}_0\|.$$

Displacements stored by the solver:

$$\mathbf{u}_A, \mathbf{u}_B, \quad \mathbf{X}_A = \mathbf{X}_A^{eq} + \mathbf{u}_A, \quad \mathbf{X}_B = \mathbf{X}_B^{eq} + \mathbf{u}_B.$$

Current (deformed) chord in space:

$$\mathbf{R} = \mathbf{X}_B - \mathbf{X}_A = \mathbf{R}_0 + (\mathbf{u}_B - \mathbf{u}_A), \quad L = \|\mathbf{R}\|, \quad \hat{\mathbf{R}} = \mathbf{R}/L. \quad (1)$$

What we previously used:

$$\mathbf{r}_{old} = \mathbf{u}_B - \mathbf{u}_A, \quad \hat{\mathbf{r}}_{old} = \mathbf{r}_{old}/\|\mathbf{r}_{old}\|.$$

Why the professor's counterexample works. If the reference bond is horizontal ($\mathbf{R}_0 = \hat{\mathbf{i}}$) but only B moves up ($\mathbf{u}_B = \hat{\mathbf{j}}$), then $\mathbf{R} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ (45°), while $\mathbf{r}_{old} = \hat{\mathbf{j}}$ (90°). A geometric spring must push/pull along $\hat{\mathbf{R}}$, not along $\hat{\mathbf{r}}_{old}$.

3 Corrected spring force (geometric)

Define the scalar extension/compression

$$s = L - L_0. \quad (2)$$

One consistent exponential analogue of Hooke's law is

$$\mathbf{F}_{A \leftarrow B} = k(e^{\alpha s} - 1)\hat{\mathbf{R}}. \quad (3)$$

Notes:

- For small $|s|$, $e^{\alpha s} - 1 \approx \alpha s$, so $F \approx (k\alpha)s$.
- If $s < 0$ (compression), then $e^{\alpha s} - 1 < 0$ and the force reverses direction, as expected.
- The direction is always along the current chord $\hat{\mathbf{R}}$ (geometrically consistent).

4 What must change in the write-up

In `papers/technical/spring_force_11x11.pdf`, the spring section should be revised so that:

- the vector used for direction is $\mathbf{R}$ in Eq. (1), not $\mathbf{u}_B - \mathbf{u}_A$;
- the scalar “stretch” entering the exponential is $s = L - L_0$ in Eq. (2) (or an equivalent measure), not $d = \|\mathbf{u}_B - \mathbf{u}_A\|$;
- nearest-neighbor bonds use $L_0 = \Delta$ and diagonal bonds use $L_0 = \Delta\sqrt{2}$, where Δ is the grid spacing.

5 What must change in the code (minimal, concrete)

Current behavior (in `src/lattice_simulation_11x11.jl`) computes the spring direction from `pos2-pos1`, where `pos` is the displacement state. To match Eq. (3) we must include the reference chord $\mathbf{R}_0$ for each neighbor link.

Minimal implementation strategy:

- In the neighbor loop (where we know whether a link is left/right/up/down/diagonal), define $\mathbf{R}_0$ from the neighbor offset:

$$\mathbf{R}_0 = \begin{cases} [\pm\Delta, 0]^T & \text{horizontal,} \\ [0, \pm\Delta]^T & \text{vertical,} \\ [\pm\Delta, \pm\Delta]^T & \text{diagonal,} \end{cases} \quad L_0 = \|\mathbf{R}_0\|.$$

- Compute $\mathbf{R} = \mathbf{R}_0 + (\mathbf{u}_B - \mathbf{u}_A)$, then $s = \|\mathbf{R}\| - L_0$, then apply Eq. (3).
- Replace the current call to `spring_force_2d(pos.k, pos_neighbor, k, alpha)` with a new routine such as `spring_force_2d_geometric(uA,uB,R0,k,alpha)`.

Also update:

- the spring potential-energy function to use s and L_0 if you want energy checks consistent with the new dynamics;
- any text in the manuscript that claims “force acts along the line connecting the masses” (this becomes true again once Eq. (3) is used).

6 Suggested discussion points for the meeting (what to show)

- The one-line correction: “use the deformed chord $\mathbf{R} = \mathbf{R}_0 + (\mathbf{u}_B - \mathbf{u}_A)$ for direction, and $s = \|\mathbf{R}\| - L_0$ for the scalar input.”
- The professor’s (4,4)–(5,4) example and the 45° chord.

- A clear statement of what changes in simulation behavior: shear no longer produces a force direction unrelated to the actual spring axis.
- Clarify whether the model should allow compression ($s < 0$) with the same exponential law or whether you prefer a tension-only/contact-like behavior; Eq. (3) allows both.